NAME :D.G.RAGHAVI
REG NO: 192521173

COURSE CODE : CSA1013
COURSE NAME : SOFTWARE ENGINEERING

ASSESSMENT TOOL - 01

Title: AI-Based Smart Construction Project Monitoring and Risk Assessment Platform

Problem Analysis

The construction industry faces several challenges, including project delays, budget overruns, safety risks, and inefficient resource management. Most construction companies rely on manual inspections and periodic reports, making it difficult to monitor project progress in real time. This often leads to delayed identification of safety violations, equipment failures, and schedule deviations. Since information is collected manually, decision-making becomes slow and inaccurate. As construction projects become larger and more complex, traditional monitoring methods are no longer sufficient. Therefore, there is a need for an AI-based platform that can continuously monitor construction sites, analyze risks, predict delays, and improve overall project efficiency.

Stakeholder Analysis

Stakeholder	**Role**	**Benefits**
Project Manager	Monitors project progress and schedules	Better planning and timely decision-making
Site Engineers	Supervise construction activities	Real-time progress updates and issue detection
Safety Officers	Monitor worker safety	AI-based safety violation detection
Construction Workers	Perform site operations	Improved workplace safety
Clients	Track project status	Greater transparency and timely completion
Company Management	Manage resources and costs	Reduced project risks and improved profitability

Current System Analysis

The existing construction monitoring system is primarily manual and has several limitations.

Existing Process

- Site inspections are performed manually.
- Progress reports are prepared periodically.
- Safety violations are identified by human observation.
- Equipment usage is tracked manually.
- Project risks are assessed after problems occur.

Limitations

- Lack of real-time monitoring.
- Delayed identification of project risks.
- Human errors in reporting.
- Poor communication among stakeholders.
- Inefficient resource allocation.
- Higher project costs due to delays.
- Limited prediction of future risks.

Proposed Software Solution

The proposed solution is an **AI-Based Smart Construction Project Monitoring and Risk Assessment Platform** that integrates Artificial Intelligence, IoT sensors, drones, cameras, BIM (Building Information Modeling), and cloud computing.

Key Features

- Real-time construction progress monitoring.
- AI-based safety violation detection.
- Delay prediction using machine learning.
- Resource allocation optimization.
- Equipment tracking and utilization analysis.
- Risk assessment dashboard.
- Automated project reports.
- Cloud-based data storage and access.

- Alerts and notifications for critical issues.
- Interactive performance dashboard.

Technologies Used

- Artificial Intelligence (AI)
- Machine Learning
- Internet of Things (IoT)
- Drone Imaging
- Building Information Modeling (BIM)
- Cloud Computing
- Web Dashboard
- Database Management System (MySQL)

SDLC / Framework Justification

Selected SDLC Model: Agile Model

The Agile Software Development Life Cycle is suitable because construction project requirements may change frequently based on client needs, safety regulations, and project progress.

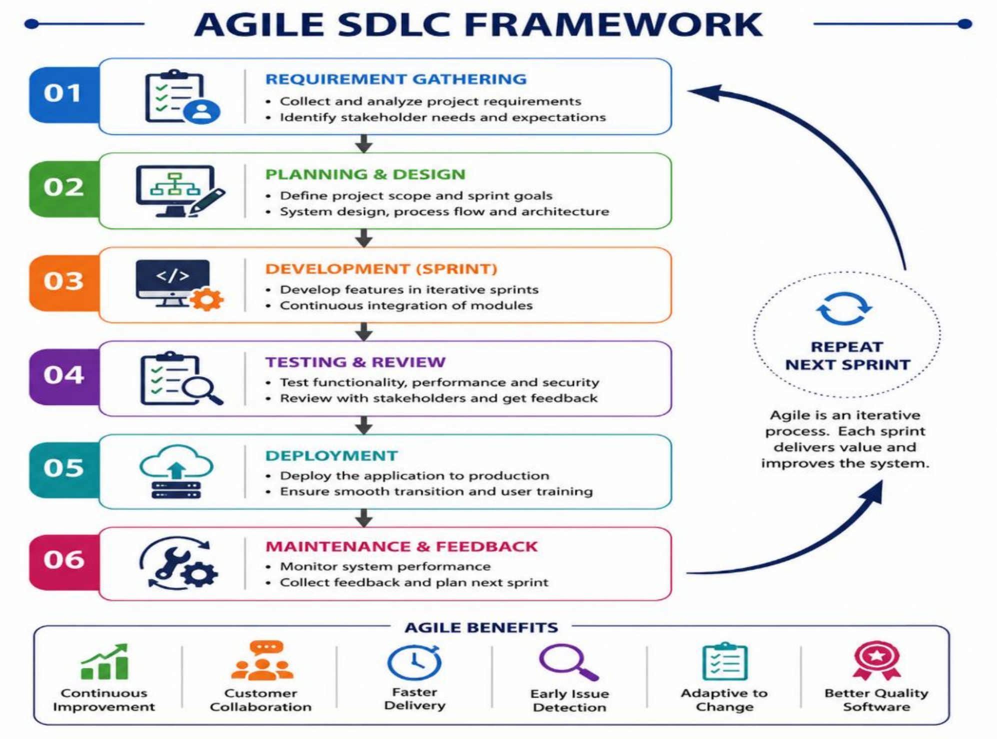
Reasons for Choosing Agile

- Supports continuous development.
- Allows frequent client feedback.
- Easy to update AI models.
- Faster delivery of software modules.
- Better risk management.
- Suitable for cloud-based applications.
- Continuous testing improves software quality.

Agile Phases

1. Requirement Gathering
2. System Design
3. Development
4. Testing

5. Deployment
6. Maintenance



Expected Outcomes and Limitations

Expected Outcomes

- Reduced project delays.
- Improved worker safety.
- Better resource utilization.
- Lower project costs.
- Accurate real-time monitoring.
- Faster decision-making.
- Increased productivity.
- Higher project success rate.
- Improved communication among stakeholders.
- Enhanced risk management.

Limitations

- High initial implementation cost.

- Requires reliable internet connectivity.
- AI predictions depend on quality data.
- IoT sensors require regular maintenance.
- Privacy concerns with continuous surveillance.
- Training is required for construction staff.
- Integration with existing systems may be complex.
- Large data storage and cloud infrastructure are needed.

Conclusion

The AI-Based Smart Construction Project Monitoring and Risk Assessment Platform provides a modern solution to overcome the limitations of traditional construction management. By combining AI, IoT, drones, BIM, and cloud technologies, the system enables real-time monitoring, predictive risk analysis, and intelligent decision-making. This leads to improved safety, reduced delays, optimized resource utilization, and overall better project performance, making construction projects more efficient and successful.